

UNIVERSITY OF MALAYA

EXAMINATION FOR THE DEGREE OF MASTER OF DATA SCIENCE

ACADEMIC SESSION 2023/2024 : SEMESTER II

WQD7006 : Machine Learning for Data Science

June 2024

Time: 1 hour and 30 minutes

INSTRUCTIONS TO CANDIDATES :

Answer **ALL** questions. (30 marks)

(This question paper consists of 4 questions on 4 pages)

Alternative assessment 1 - Online assessment/30 marks – 1.5 hours

Submission through Spectrum. Specific instructions to be given on Spectrum as well.

Question 1: 5 marks

1. You are given that:

30 emails out of 74 are spam

51 emails out of those 74 contain the word "corona"

20 emails containing the word "corona" have been marked as spam

(a) What is the probability that the latest received email is a spam message, given that it contains the word "corona"?

(2 marks)

(b) Now, you are given that:

25 emails out of the total contain the word "virus"

24 emails out of those have been marked as spam

What is the probability that an email is spam, given that it contains both "corona" and "virus"?

(3 marks)

Question 2: 6 marks

2. Patient information: gender (Boolean), age group (under 20, 20-35, 35-55, over 55), ethnicity (Malay, Chinese, Indian, Others), weight (a numerical value), height (numerical value), TB status (yes, no). All patient records are maintained to determine the type of treatment to be provided.

Answer the questions with regards to Logistic Regression.

(a) What type of problem is this? Why? (1 marks)

(b) How to cater for categorical data so that they can be used as features in the model? (2 marks)

(c) Explain what possibly happens during training and testing? (2 marks)

(d) Do you need to use regularization in this problem? Justify your answer for either yes/ no. (1 mark)

Question 3: 10 marks

3. The given dataset will be utilized to construct a decision tree to predict whether a customer will qualify for a special marketing offer (Yes or No), based on three attributes, loyalty status (Gold, Silver, or Bronze), frequency of purchases as a continuous variable, and participation in recent marketing campaigns (Yes or No).

The decision tree will analyse variable interactions to identify customers eligible for marketing offers. It will be trained using existing data to predict offer eligibility based on loyalty status, purchase frequency, and campaign involvement. The model's effectiveness will be evaluated by its prediction accuracy across diverse customer profiles.

Loyalty Status	Recent Purchase	Offer Eligible
Bronze	No	No
Bronze	Yes	Yes
Silver	No	No
Silver	Yes	Yes
Gold	No	Yes
Gold	Yes	Yes
Bronze	No	No
Bronze	Yes	Yes
Silver	No	No
Silver	Yes	Yes

Answer the following questions for the given case study.

- a. What is the entropy of the attribute "Offer Eligible" (2 marks)
- b. What is the entropy of the attribute " Offer Eligible " when considering the subset of data where the Recent Purchase is "Yes" (Offer Eligible| Recent Purchase)? (3 Marks)
- c. What is the entropy of the attribute " Offer Eligible " when considering the subset of data where the Loyalty Status is "Gold" (Offer Eligible| Loyalty Status)? (3 Marks)
- d. Draw the complete decision tree that would be learned from this dataset. You do not need to show any calculations. (2 Marks)

Question 4: 9 marks

4. Sample dataset for the education case study on clustering students based on their learning histories.

Student ID	Age	Gender	Mathematics	Reading	Writing	Learning Style	Cluster
1	10	Male	Proficient	Basic	Basic	Visual	2
2	12	Female	Advanced	Advanced	Proficient	Auditory	1
3	11	Male	Basic	Proficient	Basic	Kinaesthetic	3
4	13	Female	Advanced	Advanced	Advanced	Visual	1
5	10	Male	Proficient	Basic	Proficient	Auditory	1

Answer the following questions for the given case study.

- Calculate the Euclidean distance between student ID 2 and student ID 5 in the given education dataset. (2 Marks)
- Calculate the Manhattan distance between student ID 2 and student ID 5 in the given education dataset. (3 Marks)
- Explain the steps involved in the K-means clustering algorithm, and how it is used to cluster students based on their learning histories in the given education dataset? (4 Marks)

END